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Device and method for strengthening ball joint mounting orientation to a double shear configuration

Abstract

A support device is configured for strengthening a single shear mounting configuration of a ball joint affixed on a suspension knuckle at a first location. The support device includes a support bracket with at least one attachment member and a support mechanism. The at least one attachment member is configured for attaching the support bracket to the suspension knuckle approximate a spherical bearing attached to the suspension knuckle at the first location. The support mechanism is affixed to the at least one attachment member. The support mechanism is configured to support the spherical bearing at a second location on the spherical bearing. Wherein, the support bracket is configured to provide a double shear mounting configuration of the spherical bearing on the suspension knuckle via the combined support from the first location in combination with the second location on the spherical bearing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims benefit to U.S. Provisional Patent Application No. 63/604,446 filed on Nov. 30, 2023, entitled Device and Method for Strengthening Ball Joint Mounting Orientation to a Double Shear Configuration, which is incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to ball joints or spherical bearings for vehicles. More specifically, the present disclosure relates to a device and method for strengthening ball joint mounting orientation to a double shear mounting configuration with a spherical bearing.

BACKGROUND

(2) Generally speaking, in an automobile, ball joints are spherical bearings that connect the suspension components to the suspension knuckles (like a steering knuckle, spindle, the like, etc.) and are used on virtually every automobile made. Ball joints functions resemble the ball-and-socket joints found in most animals. A ball joint consists of a bearing stud and socket enclosed in a casing. Typically, the bearing stud, socket and casing are all made of steel. The bearing stud is tapered and threaded, and fits into a tapered hole in the suspension knuckle. A protective encasing may be used to prevent dirt from getting into the joint assembly. Usually, this protective casing may be a rubber-like boot that allows movement and expansion of lubricant.

(3) One problem associated with ball joints used to connect the suspension components to the suspension knuckle of vehicles is the shear forces or loads applied to such ball joints. Because the ball joints have a single mounting point on the suspension knuckle and the suspension component, these ball joints are mounted in a single shear configuration. Namely, the shear forces or loads applied on these ball joints are applied to the bearing stud threaded into the tapered hole in the suspension knuckle, as that is the single mounting point to the suspension knuckle. In addition, these shear forces and loads are required to be carried from the bearing stud to the socket mounted in the suspension component in a single mounting point. These shear forces or loads applied to such ball joints require the ball joints and associated hardware to be manufactured to withstand such shear forces or loads which leads to added costs, bulk and weight to the vehicle. In addition, these shear forces or loads, especially in off-road vehicles and environments, often result in ball joints regularly wearing down, failing and/or breaking, thereby, requiring such ball joints to be replaced far more often than desired. As such, there is clearly a need to provide a stronger and more stable mounting configuration for such ball joints, especially on off-road vehicles, like UTVs, Side-by-Sides, recreational off-road vehicles, the like, etc.

(4) The instant disclosure may be designed to address at least certain aspects of the problems or needs discussed above by providing a device and/or method for strengthening ball joint mounting orientation to a double shear mounting configuration with a spherical bearing.

SUMMARY

(5) The present disclosure may solve the aforementioned limitations of the currently available ball joints mounting configurations on suspension knuckles of vehicles, by providing a support device configured for strengthening a single shear mounting configuration of a ball joint affixed on a suspension knuckle at a first location. The support device may include a support bracket with at least one attachment member and a support mechanism. The at least one attachment member may be configured for attaching the support bracket to the suspension knuckle approximate a spherical bearing. The support mechanism may be affixed to the at least one attachment member. The support mechanism may be configured to support the spherical bearing at a second location on the spherical bearing. Wherein, the support bracket may be configured to provide a double shear mounting configuration of the spherical bearing on the suspension knuckle via the combined support from the first location in combination with the second location on the spherical bearing.

(6) One feature of the disclosed support device may be that the provided double shear mounting configuration can be designed or configured to create a stronger and more stable mounting configuration from a ball joint to a spherical bearing on the suspension knuckle.

(7) Another feature of the disclosed support device may be that it can be designed or configured to modify an existing ball joint assembly from the single shear mounting configuration to the double shear mounting configuration with a spherical bearing. Wherein, the support device may be

designed or configured to convert an OEM suspension pivot ball joint between a suspension component and the suspension knuckle from the single shear configuration to the double shear mounting configuration with a spherical bearing.

(8) Another feature of the disclosed support device may be that it can be designed or configured to greatly strengthen the single shear mounting configuration of the OEM suspension pivot ball joint and reduce stress on hardware.

(9) In select embodiments of the disclosed support device, the at least one attachment member may include at least one attachment flange. The at least one attachment flange may be configured and shaped for mounting the support bracket to the suspension knuckle. In select embodiments, the at least one attachment member may include two of the attachment flanges. Each of the two attachment flanges may be configured and shaped for mounting the support bracket to the suspension knuckle.

(10) In select embodiments, the disclosed support device may further include hardware, welding, or a combination thereof configured for mounting each of the attachment flanges of the support bracket to the suspension knuckle.

(11) In select embodiments of the disclosed support device, each of the at least one attachment flanges may include an attachment hole therethrough. Each of the attachment holes may be configured for receiving a mechanical fastener therethrough for mounting the support bracket to the suspension knuckle. In select embodiments, the at least one attachment member may include two attachment flanges, a first attachment flange with a first attachment hole therethrough and a second attachment flange with a second attachment hole therethrough. Wherein, the first attachment hole on the first attachment flange may be configured to mount the support bracket to an existing first threaded hole on the suspension knuckle via a first bolt, and the second attachment hole on the second attachment flange may be configured to mount the support bracket to an existing second threaded hole on the suspension knuckle via a second bolt.

(12) In select embodiments, the support device may include a connecting member affixed between the two attachment flanges. The connecting member may be configured for connecting the support mechanism with the two attachment flanges. In select embodiments, the connecting member may include a cutout. The cutout may be configured to fit around a suspension member connected to the suspension knuckle.

(13) In select embodiments of the support device, the support mechanism may include a support plate connected at an angle to the at least one attachment member. In select embodiments, the support plate may include side members configured to provide rigidity between the support plate and the at least one attachment member.

(14) In select embodiments of the support device, the support mechanism may include a support hole. The support hole may be configured to receive a top portion of a stud of the spherical bearing. In select embodiments, the support mechanism may further include a nut seat connected with the support hole. The nut seat may be sized and shaped to support a nut threaded onto the top portion of the stud of the spherical bearing.

(15) In another aspect, the instant disclosure embraces a suspension knuckle. The suspension knuckle may generally include the support device for strengthening ball joint mounting orientation to a double shear mounting configuration with a spherical bearing in any of the embodiments and/or combination of embodiments shown and/or described herein. The suspension knuckle may generally include a spherical bearing affixed to the suspension knuckle at a first location. The support device may include the support bracket mounted or affixed on the suspension knuckle. The support bracket may include at least one attachment member attached to the suspension knuckle approximate the spherical bearing, and a support mechanism affixed to the at least one attachment member. The support mechanism may be configured to support the spherical bearing at a second location on the spherical bearing.

(16) One feature of the disclosed suspension knuckle may be that the support bracket is configured

to provide a double shear mounting configuration of the spherical bearing on the suspension knuckle via the combined support from the first location in combination with the second location on the spherical bearing.

(17) Another feature of the disclosed suspension knuckle may be that the double shear mounting configuration of the spherical bearing may be designed or configured to create a stronger and more stable mounting configuration for the spherical bearing on the suspension knuckle.

(18) In select embodiments of the disclosed suspension knuckle, the at least one attachment member may include two attachment flanges. Each of the two attachment flanges may be configured and shaped for mounting the support bracket to the suspension knuckle.

(19) Another feature of the disclosed suspension knuckle may be that the support device can further include hardware, welding, or a combination thereof configured for mounting each of the attachment flanges of the support bracket to the suspension knuckle.

(20) In select embodiments of the disclosed suspension knuckle, the support device may further include a connecting member affixed between the two attachment flanges. The connecting member may be configured for connecting the support mechanism with the two attachment flanges.

(21) In select embodiments of the disclosed suspension knuckle, the support mechanism may include a support plate connected at an angle to the at least one attachment member. In select embodiments, the support plate may include side members configured to provide rigidity between the support plate and the at least one attachment member.

(22) Another feature of the disclosed suspension knuckle may be that the support mechanism can include a support hole configured to receive a top portion of a stud of the spherical bearing;

(23) Another feature of the disclosed suspension knuckle may be that the support mechanism can include a nut seat connected with the support hole. The nut seat may be sized and shaped to support a nut threaded onto the top portion of the stud of the spherical bearing.

(24) In select embodiments of the disclosed suspension knuckle, the two attachment flanges may include a first attachment flange with a first attachment hole therethrough and a second attachment flange with a second attachment hole therethrough. Wherein, the first attachment hole on the first attachment flange may be configured to mount the support bracket to an existing first threaded hole on the suspension knuckle via a first bolt, and the second attachment hole on the second attachment flange may be configured to mount the support bracket to an existing second threaded hole on the suspension knuckle via a second bolt.

(25) In another aspect, the instant disclosure embraces a method for strengthening a single shear mounting configuration of a ball joint affixed on a suspension knuckle at a first location. The method may generally include the support device for strengthening ball joint mounting orientation to a double shear mounting configuration in any of the embodiments and/or combination of embodiments shown and/or described herein. The method may first include replacing the ball joint with a spherical bearing attached to the suspension knuckle at the first location. The method may then generally include providing the disclosed support bracket in any of the embodiments and/or combination of embodiments shown and/or described herein. As such, the support bracket provided by the disclosed method may generally include at least one attachment member configured for attaching the support bracket to the suspension knuckle approximate a spherical bearing, and a support mechanism affixed to the at least one attachment member. The support mechanism may be configured to support the spherical bearing at a second location on the spherical bearing. With this provided support device, the method may further include attaching the support bracket to the suspension knuckle whereby the support mechanism supports the spherical bearing at the second location on the spherical bearing and provides a double shear mounting configuration of the spherical bearing.

(26) The foregoing illustrative summary, as well as other exemplary objectives and/or advantages of the disclosure, and the manner in which the same are accomplished, are further explained within the following detailed description and its accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will be better understood by reading the Detailed Description with reference to the accompanying drawings, which are not necessarily drawn to scale, and in which like reference numerals denote similar structure and refer to like elements throughout, and in which:
- (2) FIG. 1 is a perspective view of a suspension knuckle according to the prior art with a ball joint stud mounted thereon through a threaded hole in the suspension knuckle;
- (3) FIG. 2 is a perspective view of the device for strengthening ball joint mounting orientation to a double shear mounting configuration with a spherical bearing according to select embodiments of the instant disclosure;
- (4) FIG. 3 is another perspective view of the device from FIG. 2;
- (5) FIG. 4 is a top view of the device from FIG. 2;
- (6) FIG. 5 is a front view of the device from FIG. 2;
- (7) FIG. 6 is a cross-sectional view of the device from FIG. 2 taken from the cross-sectional angle shown in FIG. 5;
- (8) FIG. 7 is a perspective view of a spherical bearing mounted in a double shear mounting configuration between a suspension knuckle and a suspension component with the device from FIG. 2 according to select embodiments of the instant disclosure;
- (9) FIG. 8 is a partially disassembled perspective view of the spherical bearing mounted in a double shear mounting configuration of FIG. 7 with the device from FIG. 2; and
- (10) FIG. 9 is a cross-sectional perspective view of the spherical bearing mounted in a double shear mounting configuration of FIG. 7 with the device from FIG. 2;
- (11) FIG. 10 is partially disassembled cross-sectional perspective view of the spherical bearing mounted in a double shear mounting configuration of FIG. 7 with the device from FIG. 2;
- (12) FIG. 11 is a perspective view of a spherical bearing mounted in a double shear mounting configuration between a suspension knuckle and a suspension component with the device from FIG. 2 according to select embodiments of the instant disclosure; and
- (13) FIG. 12 is a flow diagram of the method for strengthening a single shear mounting configuration of a ball joint affixed on a suspension knuckle according to select embodiments of the instant disclosure.
- (14) It is to be noted that the drawings presented are intended solely for the purpose of illustration and that they are, therefore, neither desired nor intended to limit the disclosure to any or all of the exact details of construction shown, except insofar as they may be deemed essential to the claimed disclosure.

DETAILED DESCRIPTION

(15) Referring first to FIG. 1, a perspective view of prior art suspension knuckle 1 according to the prior art with prior art ball joint stud 2 mounted thereon through prior art mounting hole 3 in prior art suspension knuckle 1 is shown. As shown, prior to the instant disclosure, prior art ball joint 4 was mounted with prior art single shear mounting configuration 5 to prior art suspension knuckle 1. The instant disclosure may be designed to address at least certain aspects of the problems or needs of such prior art single shear mounting configuration 5 by providing device 10 and/or method 100 for strengthening such ball joint 4 mounting orientations to double shear mounting configuration 12.

(16) Referring now to FIGS. 2-12, in describing the exemplary embodiments of the present disclosure, specific terminology is employed for the sake of clarity. The present disclosure, however, is not intended to be limited to the specific terminology so selected, and it is to be understood that each specific element includes all technical equivalents that operate in a similar

manner to accomplish similar functions. Embodiments of the claims may, however, be embodied in many different forms and should not be construed to be limited to the embodiments set forth herein. The examples set forth herein are non-limiting examples and are merely examples among other possible examples.

(17) Referring to FIGS. 2-11, the present disclosure may solve the aforementioned limitations of the currently available ball joint vehicles and their mounting configurations between suspension knuckles and suspension components, by providing the disclosed device **10** and/or method **100** configured for strengthening such ball joint mounting orientation to double shear mounting configuration **12** with spherical bearing **13**.

(18) Device **10** for strengthening such ball joint mounting orientation to double shear mounting configuration **12** may generally include support bracket **18** with at least one attachment member **20** and a support mechanism **22**. The at least one attachment member **20** may be configured for attaching the support bracket to the suspension knuckle **14** approximate spherical bearing **13**. The support mechanism **22** may be affixed to the at least one attachment member **20**. The support mechanism **22** may be configured to support the spherical bearing **13** at second location **24** on the spherical bearing **13**. Wherein, as shown in the Figures, support bracket **18** may be configured to provide double shear mounting configuration **12** of spherical bearing **13** on suspension knuckle **14** via the combined support from first location **16** in combination with second location **24** on spherical bearing **13**.

(19) As used herein, double shear may be defined as being able to simultaneously support an object on two sides. In this case, double shear may refer to a spherical bearing **13** that is mounted between two surfaces via a bolt or piece of hardware holding the assembly together.

(20) As best shown in FIGS. 7 and 9-11, one feature of the disclosed support device **10** may be that the provided double shear mounting configuration **12** can be designed or configured to create stronger and more stable mounting configuration **26** for spherical bearing **13** on suspension knuckle **14**.

(21) When replacing the ball joint of the prior art with spherical bearing **13**, the same mounting hole **15** (see mounting hole **3** from FIG. 1) may be utilized for supporting spherical bearing **13** at first location **16**. See FIGS. 7-11). The stud **70** of the spherical bearing **13** may be inserted through same mounting hole **15**. Or, in select embodiments where same mounting hole **15** is threaded, sleeve **17** may be inserted through said same mounting hole **15**, where sleeve **17** is configured for supporting stud **70** therein said threaded same mounting hole **15**.

(22) Still referring to FIGS. 7 and 9-11, another feature of the disclosed support device **10** may be that it can be designed or configured to modify existing ball joint assembly **28** from the single shear mounting configuration of the prior art (see FIG. 1) to double shear mounting configuration **12** with spherical bearing **13**. Wherein, support device **10** may be designed or configured to convert OEM suspension pivot ball joint **30** between suspension component **32** and suspension knuckle **14** from the single shear configuration (see FIG. 1) to double shear mounting configuration **12** with spherical bearing **13**.

(23) Still referring to FIGS. 7 and 9-11, another feature of the disclosed support device **10** may be that it can be designed or configured to greatly strengthen the single shear mounting configuration of the prior art (see FIG. 1) of the OEM suspension pivot ball joint **30** and reduce stress on hardware.

(24) Referring now specifically to FIGS. 2-6, various embodiments of support device **10** are shown. Support device **10** may generally include any configuration size or shape for converting the single shear mounting configuration of the prior art (see FIG. 1) to double shear mounting configuration **12**, as shown in FIGS. 7 and 9-11. In select embodiments of the disclosed support device **10**, the at least one attachment member **20** may include at least one attachment flange **36**. The at least one attachment flange **36** may be configured and shaped for mounting support bracket **18** to suspension knuckle **14**. In select embodiments, as shown in FIGS. 2-6, the at least one

attachment member **20** may include two of the attachment flanges **36**. Each of the two attachment flanges **36** may be configured and shaped for mounting support bracket **18** to suspension knuckle **14**.

(25) As best shown in FIG. **8**, in select embodiments, the disclosed support device **10** may further include hardware, welding, or a combination thereof configured for mounting each of the attachment flanges **36** of the support bracket **18** to suspension knuckle **14**.

(26) Referring back to FIGS. **2-6**, in select embodiments of the disclosed support device **10**, each of the at least one attachment flanges **36** may include an attachment hole (**42**, **46**) therethrough. Each of the attachment holes (**42**, **46**) may be configured for receiving a mechanical fastener therethrough for mounting the support bracket **18** to the suspension knuckle **14**. In select embodiments, the at least one attachment member **20** may include two attachment flanges **36**, a first attachment flange **40** with first attachment hole **42** therethrough and a second attachment flange **44** with a second attachment hole **46** therethrough. As best shown in FIG. **8**, the first attachment hole **42** on the first attachment flange may be configured to mount the support bracket **18** to an existing first threaded hole **48** on the suspension knuckle **14** via first bolt **50**, and second attachment hole **46** on the second attachment flange **44** may be configured to mount the support bracket **18** to an existing second threaded hole **52** on the suspension knuckle **14** via a second bolt **54**. However, the instant disclosure is not so limited, and support bracket **18** may be mounted, affixed, or welded to suspension knuckle **14** by any means and at any desired location to support spherical bearing **13** at second location **24**.

(27) Still referring to FIGS. **2-6**, in select embodiments, support device **10** may include connecting member **56**. Connecting member **56** may be affixed between the two attachment flanges **36**. Connecting member **56** may be configured for connecting support mechanism **22** with the two attachment flanges **36**. In select embodiments, connecting member **56** may include cutout **58** or the like. The cutout **58** may be configured to fit around suspension member **60** connected to suspension knuckle **14**, like as best shown in FIG. **11**. As shown, the orientation of the components is a usable and functional suspension assembly. Cutout **58** may be designed or shaped to fit various sizes and shapes of suspension members **60** connected to suspension knuckle **14**. As shown in FIGS. **2-6**, in select embodiments cutout **58** on connecting member **56** may be a concave shape cutout of the top of connecting member **56**. In addition, in other select embodiments, support bracket **18** may not require cutout **58**.

(28) Referring again to FIGS. **2-6**, support bracket **18** may be designed and shaped to support spherical bearing **13** at second location **24** to create double shear mounting configuration **12** on suspension knuckle **14**. Support bracket **18** may include any component, shape or style of a support mechanism configured for support spherical bearing **13** at second location **24** to create double shear mounting configuration **12** on suspension knuckle **14**. In select embodiments of support device **10**, the support mechanism **22** may include support plate **62** connected at angle **21** (including, but not limited to, perpendicular, as shown) to the at least one attachment member **20**. In select embodiments, the support plate **62** may include side members **64** configured to provide rigidity between the support plate **62** and the at least one attachment member **20**. In select embodiments of support device **10**, the support mechanism **22** may include support hole **66**. As best shown in FIGS. **8-10**, the support hole **66** may be configured to receive a top portion **68** of a stud **70** of the spherical bearing **13**. In select embodiments, the support mechanism **22** may further include a nut seat **72** connected with support hole **66**. The nut seat **72** may be sized and shaped to support a nut **74** threaded onto the top portion **68** of the stud **70** of the spherical bearing **13**. However, the disclosure is not so limited, and support mechanism **22** may be provided with just support plate **62** having support hole **66** therethrough without nut seat **72** with nut **74** attached on top of support plate **62**. Or, support mechanism **22** may utilize other forms of retention systems or the like for supporting top portion **68** of the stud **70** of spherical bearing **13**.

(29) Referring now specifically to FIGS. **7-11**, in another aspect, the instant disclosure embraces

suspension knuckle **14**. Suspension knuckle **14** may generally include support device **10** for strengthening ball joint mounting orientation to double shear mounting configuration **12** with spherical bearing **13** in any of the embodiments and/or combination of embodiments shown and/or described herein. The suspension knuckle **14** may generally include spherical bearing **13** affixed to suspension knuckle **14** at first location **16**. The support device **10** may include support bracket **18** mounted or affixed on suspension knuckle **14**. Support bracket **18** may include at least one attachment member **20** attached to suspension knuckle **14** approximate spherical bearing **13**, and support mechanism **22** affixed to the at least one attachment member **20**. The support mechanism **22** may be configured to support the spherical bearing at second location **24** on spherical bearing **13**.

(30) Referring now to FIG. **12**, in another aspect, the instant disclosure embraces method **100** for strengthening a single shear mounting configuration of a ball joint affixed on a suspension knuckle at a first location (like as shown in FIG. **1** of the prior art). Method **100** may generally include utilizing support device **10** for strengthening ball joint mounting orientation to double shear mounting configuration **12** with spherical bearing **13** in any of the embodiments and/or combination of embodiments shown and/or described herein. Method **100** may first include step **101** of replacing the ball joint with spherical bearing **13** attached to the suspension knuckle at first location **16**. Method **100** may then generally include step **102** of providing the disclosed support bracket **18** in any of the embodiments and/or combination of embodiments shown and/or described herein. As such, the support bracket **18** provided by the disclosed method **100** may generally include at least one attachment member **20** configured for attaching support bracket **18** to the suspension knuckle **14** approximate spherical bearing **13**, and support mechanism **22** affixed to the at least one attachment member **20**. The support mechanism **22** may be configured to support the spherical bearing **13** at second location **24** on the spherical bearing **13**. With this provided support device, method **100** may further include step **104** of attaching the support bracket **18** to the suspension knuckle **14** whereby the support mechanism **22** supports the spherical bearing **13** at the second location **24** on the spherical bearing **13** and provides double shear mounting configuration **12** of spherical bearing **13**.

(31) In select embodiments of step **101** of replacing the ball joint of the prior art with spherical bearing **13**, the same mounting hole **15** (see mounting hole **3** from FIG. **1**) may be utilized for supporting spherical bearing **13** at first location **16**. See FIGS. **7-11**). The stud **70** of the spherical bearing **13** may be inserted through same mounting hole **15**. Or, in select embodiments where same mounting hole **15** is threaded, sleeve **17** may be inserted through said same mounting hole **15**, where sleeve **17** is configured for supporting stud **70** therein said threaded same mounting hole **15**.

(32) In sum, the disclosed device **10** and method **100** for strengthening a ball joint mounting orientation to double shear mounting configuration **12** may generally be support bracket **18** that can be made via machined metal or formed sheet metal. As an example, and clearly not limited thereto, support bracket **18** can be utilized to provide the ability to change the OEM ball joint configuration on an OEM suspension knuckle of a UTV/SxS or recreational offroad vehicle and mount that joint in double shear with a spherical bearing to support the joint on opposing sides vs one side (see FIGS. **7-11**).

(33) The double shear mounting or double shear mounting configuration **12** provided by the disclosed device **10**, bracket **18** or method **100** may give double the strength of the original OEM configuration at that pivot joint **13**. The attachment process can be used with bolts and hardware, or by welding to an OEM spindle.

(34) Referring to FIG. **8**, an embodiment of the support bracket **18** of support device **10** is shown being used to create a double shear mounting point **12**. Support bracket **18** may be attached to an OEM spindle with 2 bolts on the outside. The center hole, or support hole **66**, may be where the bolt passes through and supports one side of the spherical bearing (like via top portion **68** of stud **70** of spherical bearing **13**).

(35) Referring to FIGS. 9-10, a cutaway is shown of a finished assembly with the support bracket 18 of support device 10 installed. This cutaway views show the bottom side of an OEM suspension knuckle with a spherical bearing 13 on top, the double shear support bracket 18 of support device 10 is positioned above the spherical bearing 13, and a bolt or stud 70 of the spherical bearing 13 passes through the complete assembly. The double shear mounting configuration 12 is provided by support bracket 18 of support device 10 being attached to the OEM spindle with 2 bolts locating and supporting the bracket.

(36) A feature of the present disclosure may be its ability to convert an OEM suspension pivot ball joint between an arm and spindle from a single shear configuration to double shear mounting configuration 12 with spherical bearing 13.

(37) Another feature of the present disclosure may be its ability to greatly strengthen an OEM suspension configuration and reduce stress on hardware.

(38) In the specification and/or figures, typical embodiments of the disclosure have been disclosed. The present disclosure is not limited to such exemplary embodiments. The use of the term “and/or” includes any and all combinations of one or more of the associated listed items. The figures are schematic representations and so are not necessarily drawn to scale. Unless otherwise noted, specific terms have been used in a generic and descriptive sense and not for purposes of limitation.

(39) The foregoing description and drawings comprise illustrative embodiments. Having thus described exemplary embodiments, it should be noted by those skilled in the art that the within disclosures are exemplary only, and that various other alternatives, adaptations, and modifications may be made within the scope of the present disclosure. Merely listing or numbering the steps of a method in a certain order does not constitute any limitation on the order of the steps of that method. Many modifications and other embodiments will come to mind to one skilled in the art to which this disclosure pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Although specific terms may be employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation. Accordingly, the present disclosure is not limited to the specific embodiments illustrated herein but is limited only by the following claims.

Claims

1. A support device configured for strengthening a single shear mounting configuration of a suspension knuckle, the support device comprising: a spherical bearing including a stud portion and a ball portion, the stud portion including a bottom stud portion extending from a bottom of the ball portion, and a top stud portion extending from a top of the ball portion; the spherical bearing is attached to the suspension knuckle at a first location on the suspension knuckle via the bottom stud portion; a support bracket including: at least one attachment member configured for attaching the support bracket to the suspension knuckle approximate the top stud portion of the spherical bearing; and a support mechanism affixed to the at least one attachment member, the support mechanism is configured to support the spherical bearing at a second location via the top stud portion of the spherical bearing, the support mechanism including a support hole configured to receive the top stud portion of the stud portion of the spherical bearing; the support mechanism further including a nut seat connected with the support hole, the nut seat is sized and shaped to support a nut threaded onto the top stud portion of the stud portion of the spherical bearing; wherein, the support device is designed or configured to modify an existing ball joint assembly from the single shear mounting configuration to a double shear mounting configuration with the spherical bearing.

2. The support device according to claim 1, wherein the support bracket is configured to provide the double shear mounting configuration of the spherical bearing on the suspension knuckle via combined support from the first location in combination with the second location on the spherical

bearing.

3. The support device according to claim 2, wherein the double shear mounting configuration is designed or configured to create a stronger and more stable mounting configuration for the existing ball joint assembly on the suspension knuckle.

4. The support device according to claim 2 wherein, the support device is designed or configured to convert an original equipment manufacturer (OEM) suspension pivot ball joint of the existing ball joint assembly between a suspension component and the suspension knuckle from the single shear configuration to the double shear mounting configuration with the spherical bearing.

5. The support device according to claim 4 being designed or configured to greatly strengthen the single shear mounting configuration of the original equipment manufacturer (OEM) suspension pivot ball joint and reduce stress on hardware.

6. The support device according to claim 1, wherein the at least one attachment member including at least one attachment flange configured and shaped for mounting the support bracket to the suspension knuckle.

7. The support device according to claim 6, wherein the at least one attachment member includes two of the attachment flanges, each of the two attachment flanges being configured and shaped for mounting the support bracket to the suspension knuckle.

8. The support device according to claim 6 further comprising hardware, welding, or a combination thereof configured for mounting each of the at least one attachment flange of the support bracket to the suspension knuckle.

9. The support device according to claim 6, wherein each of the at least one attachment flange including an attachment hole therethrough, each attachment hole is configured for receiving a mechanical fastener therethrough for mounting the support bracket to the suspension knuckle.

10. The support device according to claim 9, wherein the at least one attachment member including two attachment flanges, a first attachment flange with a first attachment hole therethrough and a second attachment flange with a second attachment hole therethrough.

11. The support device according to claim 10, wherein the first attachment hole on the first attachment flange is configured to mount the support bracket to an existing first threaded hole on the suspension knuckle via a first bolt, and the second attachment hole on the second attachment flange is configured to mount the support bracket to an existing second threaded hole on the suspension knuckle via a second bolt.

12. The support device according to claim 10 comprising a connecting member affixed between the two attachment flanges, the connecting member is configured for connecting the support mechanism with the two attachment flanges.

13. The support device according to claim 12, wherein the connecting member including a cutout, the cutout is configured to fit around a suspension member connected to the suspension knuckle.

14. The support device according to claim 1, wherein the support mechanism including a support plate connected at an angle to the at least one attachment member, said support plate including side members configured to provide rigidity between the support plate and the at least one attachment member.

15. A suspension knuckle comprising: a spherical bearing including a stud portion and a ball portion, the stud portion including a bottom stud portion extending from a bottom of the ball portion, and a top stud portion extending from a top of the ball portion, the spherical bearing is affixed to the suspension knuckle at a first location via the bottom stud portion; a support bracket including: at least one attachment member attached to the suspension knuckle approximate the top stud portion of the spherical bearing; and a support mechanism affixed to the at least one attachment member, the support mechanism is configured to support the spherical bearing at a second location via the top stud portion of the spherical bearing, the support mechanism including a support hole configured to receive the top stud portion of the stud portion of the spherical bearing; the support mechanism further including a nut seat connected with the support hole, the

nut seat is sized and shaped to support a nut threaded onto the top stud portion of the stud portion of the spherical bearing.

16. The suspension knuckle according to claim 15, wherein: the support bracket is configured to provide a double shear mounting configuration of the spherical bearing on the suspension knuckle via combined support from the first location in combination with the second location on the spherical bearing; the double shear mounting configuration is designed or configured to create a stronger and more stable mounting configuration for the spherical bearing on the suspension knuckle; the at least one attachment member including two attachment flanges, each of the two attachment flanges being configured and shaped for mounting the support bracket to the suspension knuckle; the support bracket further includes hardware, welding, or a combination thereof configured for mounting each of the attachment flanges of the support bracket to the suspension knuckle; the support bracket further including a connecting member affixed between the two attachment flanges, the connecting member is configured for connecting the support mechanism with the two attachment flanges; the support mechanism including a support plate connected at an angle to the at least one attachment member, said support plate including side members configured to provide rigidity between the support plate and the at least one attachment member.

17. The suspension knuckle according to claim 16, wherein: the two attachment flanges including a first attachment flange with a first attachment hole therethrough and a second attachment flange with a second attachment hole therethrough, wherein the first attachment hole on the first attachment flange is configured to mount the support bracket to an existing first threaded hole on the suspension knuckle via a first bolt, and the second attachment hole on the second attachment flange is configured to mount the support bracket to an existing second threaded hole on the suspension knuckle via a second bolt.

18. A method for strengthening a single shear mounting configuration of a ball joint affixed on a suspension knuckle at a first location, the method comprising: providing a spherical bearing including a stud portion and a ball portion, the stud portion including a bottom stud portion extending from a bottom of the ball portion, and a top stud portion extending from a top of the ball portion; replacing the ball joint on the suspension knuckle with the spherical bearing attached to the suspension knuckle at the first location via the bottom stud portion; providing a support bracket including: at least one attachment member configured for attaching the support bracket to the suspension knuckle approximate the top stud portion of the spherical bearing; a support mechanism affixed to the at least one attachment member, the support mechanism is configured to support the spherical bearing at a second location via the top stud portion of the spherical bearing, the support mechanism including a support hole configured to receive the top stud portion of the stud portion of the spherical bearing; the support mechanism further including a nut seat connected with the support hole, the nut seat is sized and shaped to support a nut threaded onto the top stud portion of the stud portion of the spherical bearing; and attaching the support bracket to the suspension knuckle whereby the support mechanism supports the spherical bearing at the second location via the top stud portion of the spherical bearing and provides a double shear mounting configuration of the spherical bearing.
